Lending Under Imperfect Information

How noisy risk assessments affect lending decisions, profitability, and group outcomes

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100,000

Synthetic applicants

200

Independent noise replications

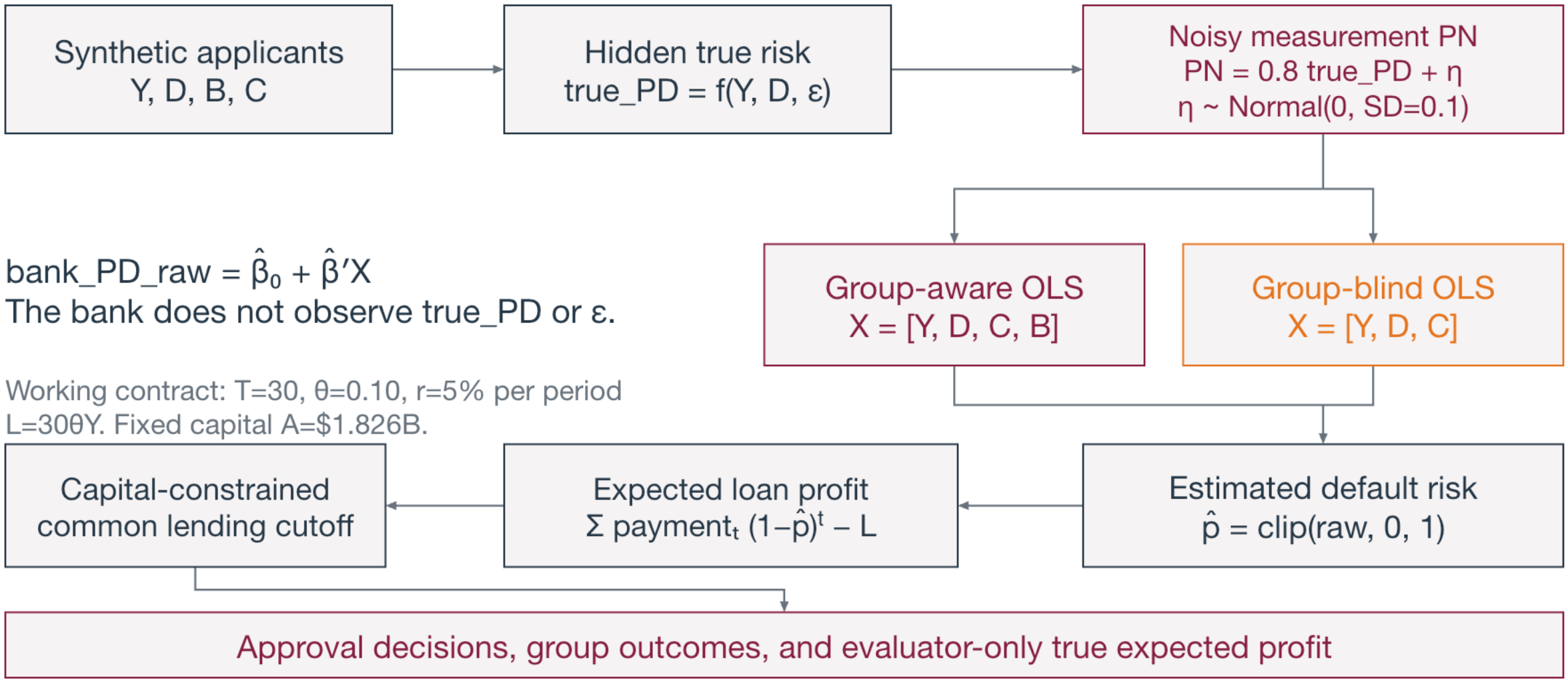
91.5%

Draws with a smaller absolute group approval gap when the bank excludes B

Noisy bank estimates can create substantial and unstable group lending disparities even when the true-risk model has no direct group effect.

Synthetic simulation, not real-world mortgage evidence. Source: Step 8B, noise_full_results.csv

From True Risk to Lending Decisions



$\text{bank_PD_raw} = \hat{\beta}_0 + \hat{\beta}'X$

The bank does not observe true_PD or ε.

Working contract: T=30, θ=0.10, r=5% per period
L=30θY. Fixed capital A=$1.826B.

Source: professor-model Steps 2–7B, saved specifications and evaluation artifacts.

How We Simulated the Lending Data

100,000 applicants

80,000 train / 20,000 evaluation

Y Annual income
D Debt-to-income ratio
B Binary group (15.07% B=1)
C Noisy credit signal
ε Hidden borrower factor

true_PD =
sigmoid(0.52 – 0.65 log Y)
× sigmoid(4 + 0.06 D + ε)

Mean true_PD 0.1422%
Noise SD 10 pp ≈ 183× scaled-signal SD

EVALUATOR-ONLY DATA PREVIEW

First five saved IDs. PD in percent, PN in raw units.

Applicant ID	B	Y ($)	D	C	PD (%)	PN raw
prof_000001	0	28,753	0.521	0.591	0.2109	0.0684
prof_000002	0	60,933	0.125	0.388	0.1295	0.0583
prof_000003	0	86,547	0.180	-0.758	0.0930	0.2353
prof_000004	1	58,043	0.128	-0.109	0.1314	-0.0432
prof_000005	0	24,706	0.257	0.985	0.2338	0.0347

Negative PN is a noisy measurement, not negative true risk.
No empirical U.S. mortgage sample or representative preview.

Source: baseline bank-view, simulation_truth.csv, training_signals.csv. Researcher-defined synthetic distributions.

Initial Bank, Group-Blind, and Oracle Results

Single synthetic noise realization, not a robustness result

Approval gap (B=1 minus B=0, pp)



True expected portfolio profit

Group-aware bank

$2.3661B

Group-blind bank

$2.3666B

True-risk oracle

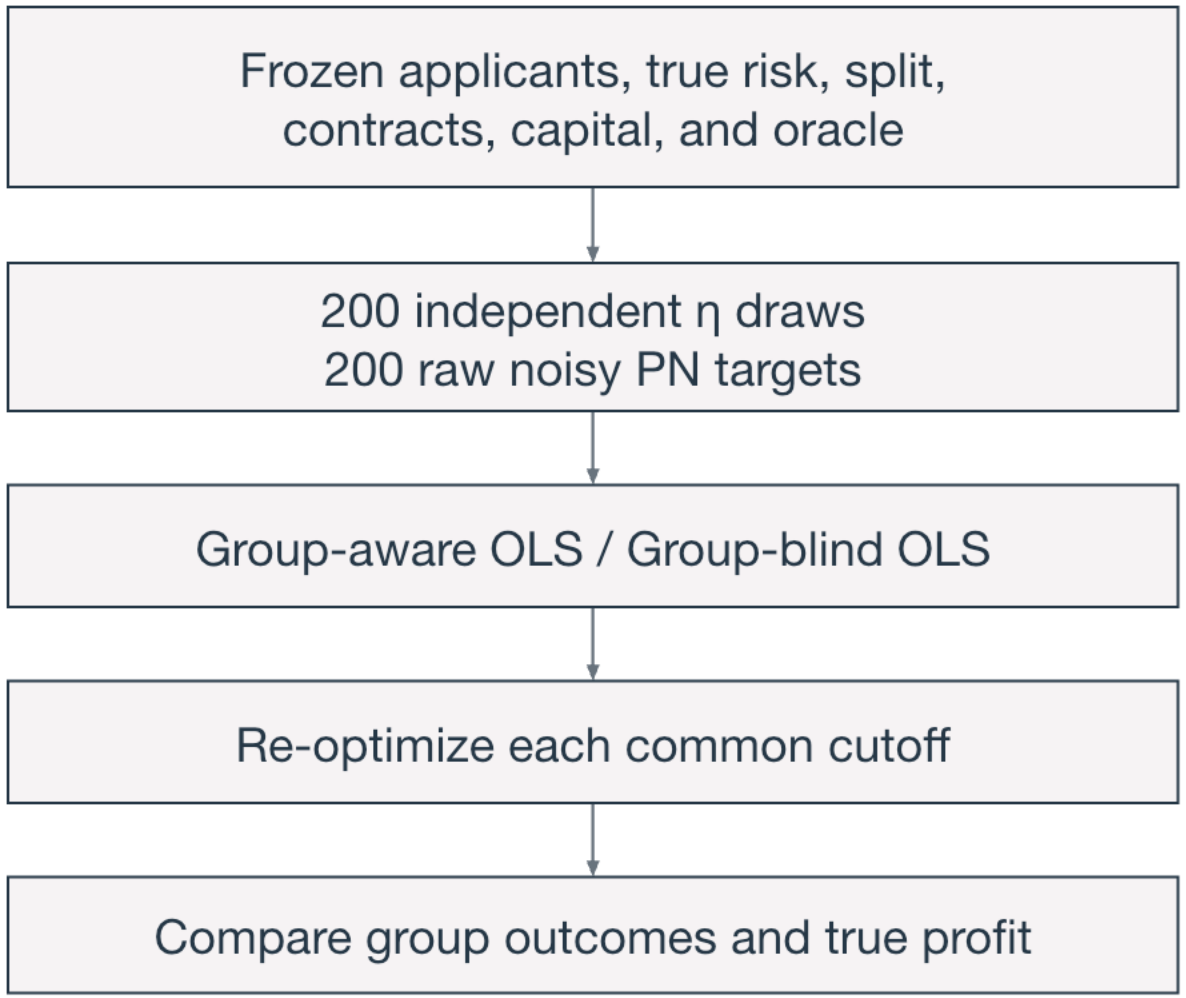
$2.3702B

Excluding B changed the original approval gap from –14.39 pp to +0.16 pp.

Different risk rankings create different feasible cutoff portfolios.
The oracle is neither a fairness rule nor an unrestricted optimality bound.

Source: group_aware_vs_blind_portfolio.csv and group_aware_vs_blind_group_audit.csv, Steps 7A–7B.

Testing Whether the Result Survives New Noise



200/200
Successful replications

400
Policy observations

0
Scientific failures

Only η changed

First three predeclared indices (gaps in pp)

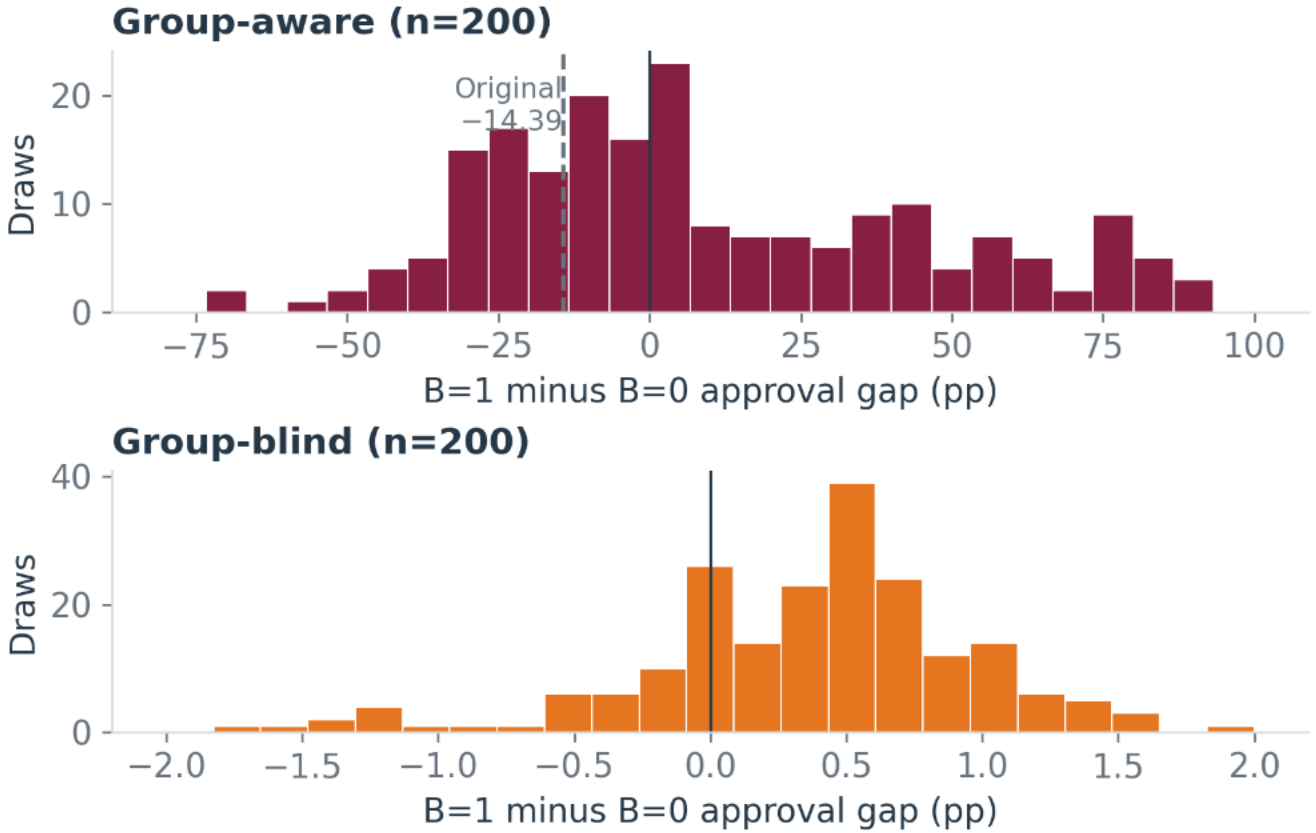
Draw	Aware	Blind	Blind–aware
0	9.975	0.754	-9.221
1	-1.331	0.976	2.307
2	18.168	0.652	-17.516

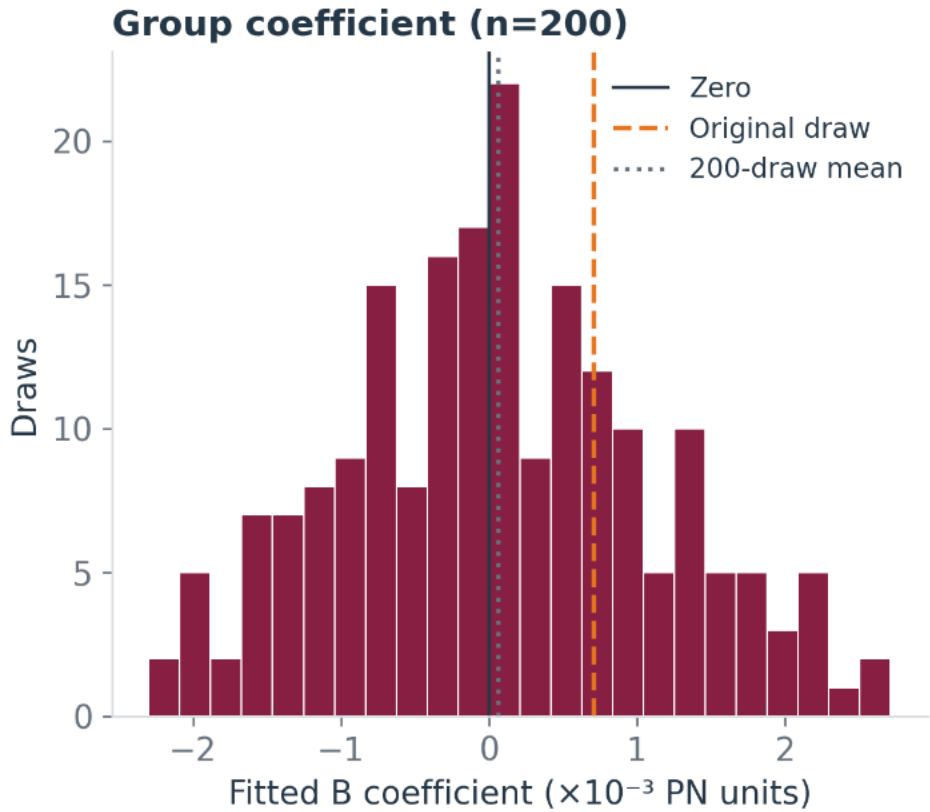
5 pilot reuses + 195 new replications

Source: Step 8B manifest, immutable replication records, noise_full_results.csv. Indices 0–199, master seed 499408.

Noisy Group-Aware Estimates Produce Unstable Gaps

The original gap changed direction across independent noise draws.





Aware mean +8.840 pp, SD 36.206 [–46.658, +82.129]
Blind mean +0.366 pp, SD 0.599 [–1.200, +1.454]

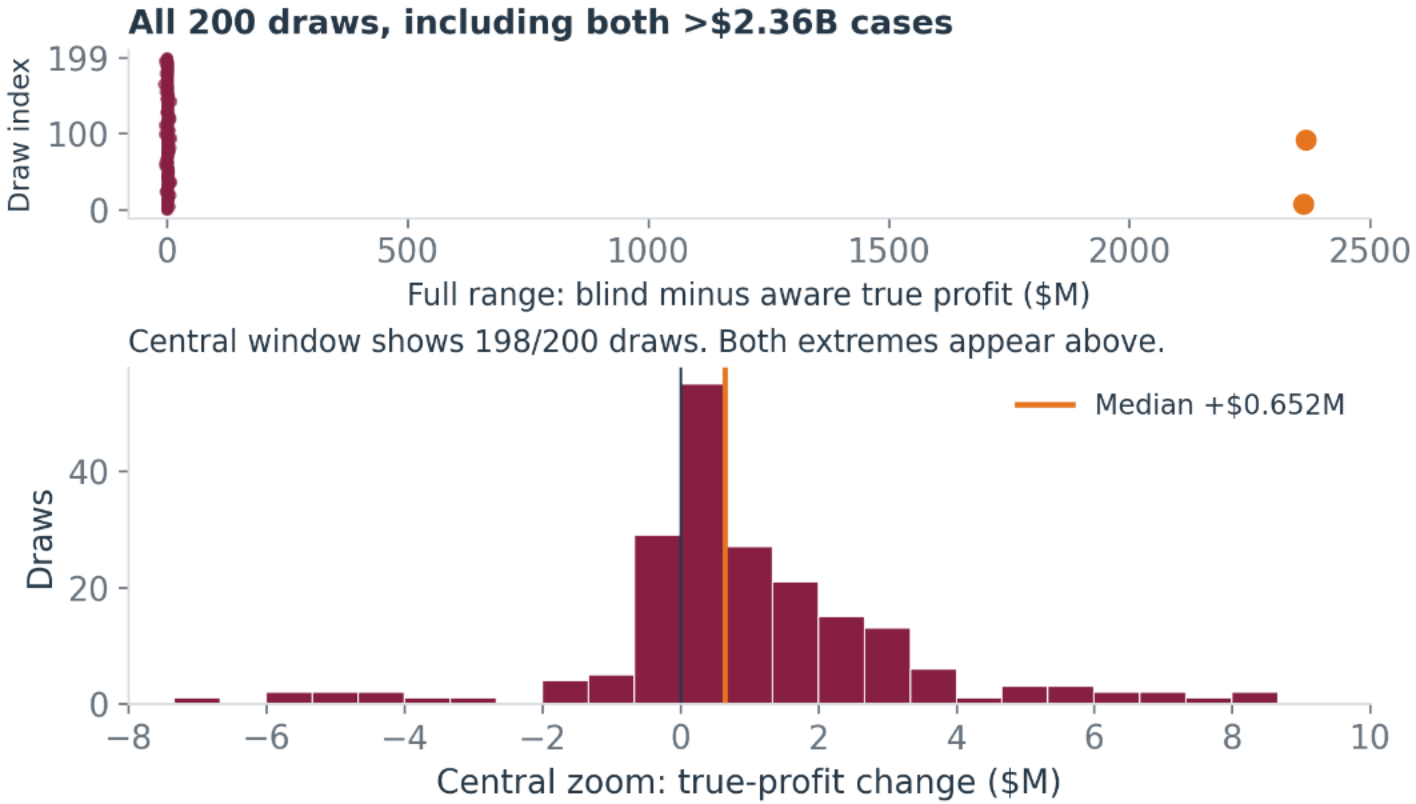
91.5% had smaller blind absolute gaps

52.5% positive / 47.5% negative B slopes
Nearly half of aware gaps were negative.

65.5% reached the original |14.39 pp| gap

Source: noise_full_results.csv, 200 draws. Brackets show p2.5–p97.5 empirical noise spreads, not causal intervals.

Profit Results Are Driven by Rare Extreme Cases



MEDIAN CHANGE

+$651,807

Mean +$24.616M

70.5% blind profit higher
23.5% lower / 6.0% equal

Two draws exceed +$2.36B each

Aware: zero loans in both
Blind: 4,744 / 3,861 loans

Reject-all: aware 13/200, blind 11/200
Zero block >90% capital: 17/200 each

Negative OLS predictions → Clip to zero → Indivisible zero-score block → Block exceeds capital → Reject-all only feasible cutoff

The mean advantage does not represent the typical replication.

Source: noise_full_results.csv. Nominal true expected profits under stylized 30-period growing payments, not realized returns.

What We Learned and What Comes Next

Three main takeaways

Noise matters

Very noisy measurements change fitted group coefficients and lending outcomes.

Group-blind usually reduced disparity

Smaller absolute gap in 91.5% of draws.
Equal outcomes are not guaranteed.

Economic interpretation needs care

Rare clipped-zero / reject-all cases dominate the mean profit difference.

Recommended next steps

01 Review noise SD 0.1 against the very small underlying true-risk variation.

02 Review clipping and the all-or-none zero-score cutoff behavior.

03 Predeclare a matched B-coefficient intervention to separate refitting effects.

04 Then predeclare alternative noise levels and independent new populations.

Next goal: isolate mechanisms before broader claims about discrimination or ML improvements.

No intentional discrimination established. Step 8B complete.
Latest documented suite: 624 passed, 4 unrelated V2 failures, 32 warnings.
Source: Step 8B audited records and current PROJECT_STATUS.md. Synthetic evidence only.